

Prime Number Dataset Analysis

Investigating Twin, Sexy, and Fibonacci Primes up to 10^6

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Introduction

Prime numbers are foundational in the world of mathematics, and have been for centuries, prompting investigations into their distribution and patterns. While the frequency of individual primes decreases as they grow larger, certain subsets of primes, such as twin (differing by 2) and sexy (differing by 6) primes appear in a different frequency in structured pairs. This report discusses the research question: **In what intervals of 100k(from 1 to 1 million) are there the most of certain types of primes, and how do their frequencies compare?**

Data Sample

To answer the research question, a public dataset of the first 1 million prime numbers "First 1 Million Prime Numbers" was downloaded and analysed from Kaggle. There were two datasets in this file - 1 containing the first 1 million prime numbers, and another containing the first 100,000 prime numbers. The script uses the file containing the first 1 million primes, processing them and resulting in a total sample of 78,498 primes.

Data Analysis Method

The analysis of the dataset was coded in Visual Studio Code, using C++. The plots were generated using Python, in the same repository. The methodology involved:

- Filtering primes less than 10^6
- Finding pairs by calculating differences (difference of 2 for twin primes, of 6 for sexy primes)
- Checking each prime against a generated set of Fibonacci numbers up to 10^6
- Aggregating counts per 100k interval into a CSV file (primestats.csv) and visualising them via a bar chart generated by Python code.

These steps were done using various functions, which were all called at the end of the code. The fact that the dataset was already ordered sped up this project and the running of the code as the analysis could be done in one pass.

To rigorously evaluate the findings, statistical uncertainties for the counted prime subsets were modeled using Poisson counting statistics ($\sigma = \sqrt{N}$). The significance Z of the difference between interval counts (N_1 and N_2) was calculated via error propagation:

$$Z = \frac{N_1 - N_2}{\sqrt{N_1 + N_2}}$$

Similarly, uncertainties for the ratio R of sexy to twin primes were propagated using:

$$\sigma_R = R\sqrt{\frac{1}{N_{sexy}} + \frac{1}{N_{twin}}}$$

Results and discussion

The quantitative analysis reveals clear patterns in the distribution of prime pairs across the 100k intervals:

- **Global Density:** Out of the first $1,000,000$ integers processed, exactly $78,498$ primes were identified, representing an overall density of approximately 7.85%
- **Twin vs. Sexy Primes:** Across the entire range below 10^6, 8,169 $\pm$ 90.4 twin pairs and 16,386 $\pm$ 128.0 sexy pairs were found, yielding a ratio of sexy to twin primes of 2.006 $\pm$ 0.027.

- **Interval Comparison:** The 0-100k interval contains the highest frequency for all examined types of primes. For example, twin pairs drop significantly by about 43% between the first interval (1,224 $\pm$ 35.0) and the last interval (697 $\pm$ 26.4), representing a statistically significant drop of 12σ. Sexy pairs experience a similar 43% drop (2,447 $\pm$ 49.5 down to 1,386 $\pm$ 37.2), representing a 17σ significance drop.
- **Fibonacci Primes:** Fibonacci primes are substantially rarer than either pair type, with only 9 $\pm$ 3 identified below one million

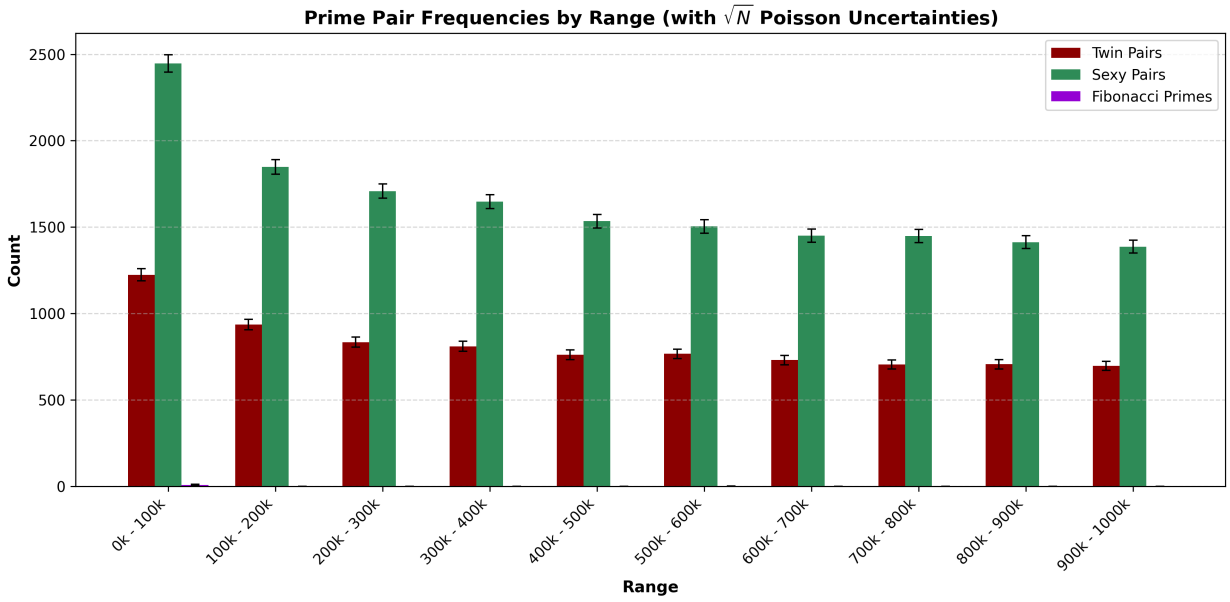


Figure 1: Distribution of prime types across 100k intervals up to 10^6.

Conclusion

In conclusion, the 0-100k interval consistently harbours the highest concentration of all evaluated prime types. Furthermore, sexy primes outnumber twin primes by a factor of roughly 2:1, while Fibonacci primes remain exceedingly sparse across the 10^6 range.

References

[1] Charles Averill, *First 1 Million Prime Numbers*, Kaggle Dataset, Available at: `https://www.kaggle.com/datasets/charlesaverill/first-100000-prime-numbers` (accessed September/October 2026).